

Lakshyaraj Singh Rao

Backend Engineer · AI Infrastructure · Full-Stack · DevOps

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· npm

SUMMARY

Backend-focused engineer building production AI infrastructure. Four published npm packages (@ykstorsorg/goldset, quickdraw, tripwire, anvil — the last with npm build provenance) and a public GitHub Marketplace Action. Patterns from a solo production AI deployment, extracted into seven open-source repos: refusal-first RAG, mid-stream guardrails, an eval harness, a streaming benchmark, a GitOps Kubernetes lab, an in-browser runtime, and an idempotent webhook-to-queue pipeline.

TECHNICAL SKILLS

Languages

TypeScript · JavaScript
Python · SQL
Rust (learning)

Backend / Data

Node.js · Express · Next.js
PostgreSQL · pgvector
Prisma 7 · Redis · BullMQ

Infra / AI

Docker · Kubernetes · Helm
Terraform · GitHub Actions · Vercel
OpenAI · Anthropic · Ollama

OPEN-SOURCE PROJECTS

Anvil [Creator](#) · github.com/ykstorm/anvil

- Idempotent webhook → BullMQ worker pipeline: HMAC-SHA256 constant-time verify, dedupe by sha256(signature + payload), retry backoff [1s, 5s, 30s, 5m], dead-letter replay via a separate consumer.
- Five design contracts proven by tests in CI (Redis-backed). Ships a Terraform module + Helm chart. Published as `ykstorsorg/anvil v0.1.0` on npm with build provenance.

TypeScript BullMQ Redis Express Terraform Helm

Goldset [Creator](#) · [Maintainer](#) · github.com/ykstorm/goldset

- Three-runner AI eval framework (goldenDataset · llmJudge · structural assertions) shipped as a public GitHub Action.
- PR comment diffs the delta on every push; structural-assertion failures block merge.
- Published as `ykstorsorg/goldset v0.2.0` on npm and listed on GitHub Marketplace (Continuous Integration).

TypeScript esbuild OpenAI Anthropic GitHub Actions

Anchor [Creator](#) · anchor-iota-ten.vercel.app/playground

- Provenance-first RAG layer with cosine-floor refusal. Returns chunks when similarity ≥ 0.30 ; returns refused: true and zero chunks below floor.
- Idempotent upsert keyed on (sourceType, sourceId) — safe to re-run without duplicates.
- Live playground renders the retrieval decision — chunks above the floor, or an explicit refusal — for any query.
- Adaptive K per query intent, with a provenance array on every response so answers are traceable to source.

Postgres + pgvector Prisma 7 Next.js 15 OpenAI embeddings

Tripwire [Creator](#) · github.com/ykstorm/tripwire

- Mid-stream LLM safety. Token-by-token rule engine aborts the response stream on rule trip.
- Sub-millisecond decision per chunk; zero buffering on happy path.
- User never sees a bad response finish rendering — abort happens before the next token reaches them.
- Published as `ykstorsorg/tripwire` on npm.

TypeScript Node async iterators streaming abort

Quickdraw [Creator](#) · github.com/ykstorm/quickdraw

- LLM streaming benchmark CLI. Captures TTFT, tokens-per-second, and dollar-per-1K tokens.
- Nightly run at 03:00 UTC across OpenAI + Anthropic; diffs vs. baseline and commits results to repo.
- Published as `ykstorsorg/quickdraw` on npm.

TypeScript OpenAI SDK Anthropic SDK GHA cron

Stackup [Creator](#) · github.com/ykstorm/stackup

- Local Kubernetes lab spinning up in under 10 minutes via single make up target.
- ArgoCD + Argo Rollouts + Grafana + Prometheus on kind cluster — production-shaped stack for learning.

Kubernetes ArgoCD Helm kind Grafana

Codecraft [Creator](#) · codecraft-ai-tau.vercel.app

- Boots a real Next.js dev server inside the browser tab via WebContainers — no backend, no local install. The whole cross-origin-isolation (COOP/COEP) boundary is what makes SharedArrayBuffer and the in-tab runtime work.

WebContainers Monaco xterm Next.js COOP/COEP

FREELANCE WORK

Homesty.ai [Solo Backend & AI Engineer](#) · Remote · Mar 2025 – Present

- Sole engineer on a live buyer-side real-estate AI: Next.js 15, Postgres + pgvector, Prisma, GPT-4o and Claude streaming, behind rate limiting, intent routing, and Sentry.
- Built the chat path around refusal-first retrieval and a mid-stream guardrail — the AI declines when it has no grounded source rather than improvising, which in real estate is a liability, not a UX choice.
- Hardened those two patterns into standalone OSS the rest of the industry can use: Anchor (provenance RAG) and Tripwire (streaming safety).

Next.js 15 Postgres + pgvector Prisma 7 GPT-4o Claude Sentry Docker

EDUCATION

Manipal University Jaipur — B.Tech, Computer Science · Graduating 2026

Relevant coursework: Distributed Systems, Database Internals, Networks, ML Foundations

RECOGNITION

- Four npm packages under @ykstorg — Anvil published with npm build provenance (signed on GitHub Actions)
- Goldset listed on GitHub Marketplace (Continuous Integration)
- Seven open-source repositories, each with a working demo and green CI
- Portfolio with live npm + GitHub activity at lakshyaraj-dev.vercel.app

Latest version: github.com/ykstorm/resume · Apache 2.0